FCM 2: Assignment 1

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1 Executive Summary

This project studies the behavior of classical nonlinear root finding and fixed point itterations through a sequence of controlled numerical experiments drawn from Problem Set 6. The emphasis is on understanding how well known iterations behave in practice when confronted with multiple roots, singularities, scaling effects, cycling, and coupled nonlinear systems

I implemented the following nonlinear solvers:

(i) Newton's method

(ii) Secant method

(iii) Regula Falsi

(iv) Steffensen's method

(v) Fixed point itteration (scalar)

(vi) Fixed point itteration (vector)

All routines use uniform stoping criteria based on iteration count, residual size, and domain validation checks.

Four experimental tasks were conducted:

1. Newton itteration on a cubic with three real roots
2. Scaling effects on nonlinear itterations
3. Root coalescing and multiplicity effects
4. Fixed point methods for non linear systems

throughout these experiments, I recorded relevant data including: convergence or failure, itteration counts, residual histories, excursion size, basins of attraction

Main observations:

1. Newton converges rapidly when initialized appropriately, but can exhibit cycling and large excursions near singularities.
2. Scaling strongly degrades convergence of derivative free methods, while Newton is largely insensitive to scaling.
3. As roots coalesce, convergence slows and becomes more sensitive to initiallization.
4. Fixed point formulations for the nonlinear system show strongly asymetric robustness and convergence behavior.

Overall, the experiments illustrate how convergence behavior depends on scaling, root structure, and the specific form of the iteration. The scalar tests exhibit the expected sensitivity to multiplicity and derivative behavior, while the system experiments show that different relaxation maps can have very different robustness, each converging to only one solution. These results are consistent with the theoretical assumptions behind the methods and highlight practical limitations that arise in numerical implementations.

2 Statement of the Problem

The goal of this project is to study the behavior of several nonlinear root finding and relaxation methods through a sequence of numerical experiments. The problems are drawn from **Problem Set 6** and are organized into four tasks, each designed to highlight different aspects of convergence, robustness, and failure modes of itterative methods

Tasks 1 and **task 2** focus on scalar root finding problems. In **Task 1**, classical methods (Newton, modified Newton, Secant, Steffensen, and Regula Falsi) are applied to functions with known roots and varying multiplicity in order to examine convergence rates and sensitivity to the multiplicity parameter.

Task 2 extends this analysis by considering the effects of scaling and root coalescence on convergence behavior, as well as the appearance of cycling and slow convergence for certain initial guesses.

Task 3 studies scalar nonlinear relaxation iterations of the form $x_{k+1} = \phi(x_k)$. The emphasis is on how fixed points, singularities, and the choice of initial data affect convergence or divergence. Representative initial conditions are used to illustrate convergence to different fixed points, as well as failure due to instability or lack of contractivity.

Task 4 considers nonlinear relaxation methods for a system of two equations. Two different relaxation maps are tested on the same system, and their convergence behavior is compared across a grid of initial conditions. The objective is to assess robustness, sensitivity to domain restrictions, and the extent to which each map converges to a particular solution of the system.

Across all tasks, the focus is on observed numerical behavior rather than algorithmic optimization. Iteration counts, convergence diagnostics, and failure cases are recorded to provide a practical picture of how these methods perform under varying problem structures and initial conditions.

3 Mathematical Description

3.1 Root finding as fixed point iteration

Many of the methods considered in this project can be written in the form

$$x_{k+1} = \phi(x_k),$$

where $\phi : \mathbb{R} \to \mathbb{R}$ is an iteration map whose fixed points coincide with the roots of a given nonlinear function f. For example, Newton's method applied to $f(x) = 0$ can be written as

$$\phi(x) = x - \frac{f(x)}{f'(x)},$$

so that any root x^* of f satisfying $f'(x^*) \neq 0$ is a fixed point of ϕ.

This reformulation allows root-finding methods to be analyzed using tools from fixed point theory and dynamical systems. In particular, convergence depends on local properties of ϕ near a fixed point rather than directly on f itself.

3.2 Local convergence and the Banach fixed point theorem

A central theoretical tool is the Banach fixed point theorem. Let (X, d) be a complete metric space and supose $\phi : X \to X$ satisfies a Lipschitz condition

$$d(\phi(x), \phi(y)) \leq L\, d(x, y), \qquad 0 \leq L < 1,$$

for all $x, y \in X$. Then ϕ has a unique fixed point $x^* \in X$, and for any initial guess $x_0 \in X$, the itterates x_k converge to x^*.

In the scalar setting, this condition reduces to

$$|\phi'(x)| \leq L < 1$$

on an interval containing the fixed point. When this holds locally near x^*, the iteration converges linearly. This provides a clear explanation for why many methods converge only when the derivative of the iteration map is sufficiently small near the root.

3.3 Order of convergence and derivative conditions

If ϕ is diferentiable at a fixed point x^*, the behavior of $\phi'(x^*)$ determines the local convergence rate:

- If $|\phi'(x^*)| < 1$, convergence is linear.
- If $\phi'(x^*) = 0$ but $\phi''(x^*) \neq 0$, convergence is typically quadratic.
- If $|\phi'(x^*)| > 1$, the fixed point is repelling and the iteration diverges.

For Newton's method applied to a simple root ($f'(x^*) \neq 0$), one obtains $\phi'(x^*) = 0$, leading to quadratic convergence. For roots of higher multiplicity, however, $\phi'(x^*) \neq 0$, and the convergence degrades to linear unless the method is modified.

This explains the behavior observed in Task 1 when standard Newton's method is applied to multiple roots and motivates the use of modified Newton iterations.

3.4 Scaling, coalescing roots, and loss of contractivity

Tasks 1 and 2 also explore how scaling and root geometry affect convergence. While scaling a function does not change its roots, it can significantly alter the derivatives appearing in an iteration map. As a result, the Lipschitz constant of ϕ may increase, reducing the size of the region in which the iteration is a contraction.

Similarly, when two roots move closer together (root coalescence), the derivative f' becomes small between the roots, and itteration maps derived from f may lose contractivity altogether. In such cases, even methods that converge rapidly for well-separated roots may exhibit slow convergence, cycling, or outright failure. These effects are observed numerically in Task 2 and are consistent with the loss of the assumptions required by the Banach fixed point theorem.

3.5 Nonlinear relaxation and dynamical systems viewpoint

In Tasks 3 and 4, the iteration maps are not derived from Newton-type corrections but are prescribed nonlinear relaxations. From a dynamical systems perspective, each iteration defines a discrete-time dynamical system, and fixed points correspond to equilibria.

The local stability of a fixed point is determined by the Jacobian of the iteration map. In the scalar case,

$$|\phi'(x^*)| < 1$$

implies that x^* is locally attracting, while $|\phi'(x^*)| > 1$ implies that it is repelling. Singularities of ϕ introduce regions where the iteration is not well defined, often leading to divergence or unbounded growth of the iterates.

For systems, the iteration takes the form

$$\mathbf{x}_{k+1} = \mathbf{G}(\mathbf{x}_k),$$

and local convergence is governed by the spectral radius of the Jacobian $DG(\mathbf{x}^*)$. If

$$\rho(DG(\mathbf{x}^*)) < 1,$$

then $\mathbf{x}^*$ is locally attracting. If $\rho(DG(\mathbf{x}^*)) > 1$, the fixed point is unstable. This framework explains why, in Task 4, each relaxation map converges reliably to only one of the two solutions of the system and fails to converge to the other.

3.6 Practical implications

While fixed point theory provides clear sufficient conditions for convergence, the numerical experiments in this project illustrate that these conditions are often violated in practice. Iterations may fail due to poor scaling, proximity to singularities, or loss of contractivity far from the fixed point. As a result, the observed behavior depends strongly on the iteration design and the choice of initial data, reinforcing the need for careful analysis when applying nonlinear iterative methods.

4 Implemented Subroutines

This project relies on a small collection of scalar and vector iteration routines used to study fixed point behavior, convergence rates, and sensitivity to problem structure and initial conditions.

Scalar fixed point iteration

`fixed_point.m` implements the basic fixed point iteration

$$x_{k+1} = \phi(x_k),$$

with user supplied iteration map ϕ. The routine:

- iterates until either $|x_{k+1} - x_k| < \texttt{tol}$ or `maxit` is reached,
- records the itterate history and residuals,
- flags nonconvergence due to divergence or stagnation.

This routine is used directly in **Task 3** to study nonlinear relaxation and basin behavior.

Newton type methods

`Mnewton.m` implements the modified Newton iteration

$$x_{k+1} = x_k - m\,\frac{f(x_k)}{f'(x_k)},$$

where m is a user specified multiplicity parameter. This allows direct comparison between:

- standard Newton ($m = 1$),
- correctly scaled Newton ($m = d$),
- under and over scaled variants ($m = d \pm 1$).

This routine is central to **Task 1**, where convergence degradation due to incorrect multiplicity is examined empirically.

Derivative free methods

Two derivative free methods are included:

- `secant.m`, implementing the standard secant iteration,
- `steffensen.m`, implementing Steffensen's acceleration method.

These routines are used to study sensitivity to initial guesses and to contrast theoretical order of convergence with observed floating point behavior.

Vector fixed point iteration

`fixed_point_vec.m` generalizes fixed point iteration to systems,

$$x_{k+1} = G(x_k), \qquad x_k \in \mathbb{R}^2,$$

with convergence measured using the residual norm

$$\|x_{k+1} - x_k\|_2.$$

This routine supports domain checks to prevent invalid iterates (e.g. square roots of negative values) and is used in **Task 4** to analyze nonlinear relaxation for systems.

Problem specific iteration maps

The functions `G1.m` and `G2.m` implement two different fixed point reformulations of the same nonlinear system from Problem 6.10. Comparing these maps highlights how algebraically equivalent systems can exhibit drastically different convergence behavior due to changes in contraction properties.

Bracketing methods

`regula_falsi.m` implements the classical false position method and is used in **Task 1** to study robustness under multiple roots and sign change assumptions.

5 Driver and Experiments

A separate driver was written for each task to organize experiments, control parameters, and collect diagnostic output. All drivers use a fixed tolerance $\texttt{tol} = 10^{-12}$ and a maximum iteration count of $\texttt{maxit} = 60$ (unless otherwise noted), and they record iteration counts, convergence status, and final residuals. Figures generated by the drivers are referenced below and collected in Appendix A.

Task 1: Newton family and related methods

The Task 1 driver applies several scalar root finding methods to functions of the form

$$f(x) = (x - \rho)^d,$$

with $\rho = 1$ and degrees $d = 1, \ldots, 6$. The following methods are tested: standard Newton's method, modified Newton with different multiplicity parameters, Steffensen's method, the Secant method, and Regula Falsi.

For each degree, the driver sweeps a small set of initial guesses on both sides of the root and records iteration counts, convergence flags, final iterates, and residual magnitudes. Additional runs examine sensitivity to incorrect multiplicity parameters. The goal is to empirically compare convergence behavior as the root multiplicity increases.

Task 2: Scaling and root structure

Task 2 focuses on the cubic polynomial

$$f(x) = x(x - \rho)(x + \rho), \qquad \rho = 1,$$

which has three real roots and stationary points where $f'(x) = 0$. The driver explores three related experiments:

- baseline convergence behavior for Newton's method across a range of initial guesses,
- the effect of scaling the function by a constant factor,
- the effect of root coalescence by allowing two roots to approach each other.

A dense sweep of initial guesses is used to identify convergence regions, cycling, and slow convergence. Summary counts of successful and failed runs are reported, and representative trajectories are plotted.

Task 3: Scalar nonlinear relaxation

In Task 3, the driver studies the scalar iteration

$$x_{k+1} = \phi(x_k), \qquad \phi(x) = x - \frac{x^2 - 3}{x^2 + 2x - 3},$$

which has two fixed points and singularities in the real line. Several initial guesses are chosen to illustrate convergence to each fixed point, sensitivity near singularities, and divergence when the iteration leaves the domain of validity.

For each run, the driver prints a detailed iteration table, including residuals and distance to the nearest fixed point, along with a final convergence summary.

Task 4: Nonlinear relaxation for systems

Task 4 considers a system of two nonlinear equations and compares two different relaxation maps, denoted G_1 and G_2. For each map, the driver performs:

- representative test runs from selected initial guesses,
- a grid sweep over initial conditions in a square domain,
- classification of convergence to each solution or failure.

Convergence is assessed using the residual $\|x - G(x)\|_2$, along with checks for non-finite iterates. Summary statistics include counts of converged and failed starts, iteration counts for successful runs, and identification of which solution each map converges to.

6 Results and Discussion

6.1 Task 1: Newton-type methods and multiplicity effects

Task 1 examines the behavior of several root-finding methods as the root multiplicity d increases, with particular emphasis on how supplying the correct multiplicity parameter affects Newton's method. The driver sweeps $d = 1, \ldots, 6$ and several initial guesses near the root $\rho = 1$.
A representative subset of the command-window output for $d = 2$ and $x_0 = 0.5$ is shown in Table 1. This behavior is typical across all tested values of d.

Method	x_0	Iter	Conv	x_{end}
Newton ($m = 1$)	0.5	19	1	0.9999990463
Newton ($m = d$)	0.5	1	1	1.0000000000
Newton ($m = d - 1$)	0.5	19	1	0.9999990463
Newton ($m = d + 1$)	0.5	19	1	1.0000009537
Steffensen	0.5	15	0	0.9999914357

Table 1: Task 1: Representative output for $d = 2$, $x_0 = 0.5$.

Several consistent patterns appear throughout Task 1:

- When the correct multiplicity is supplied ($m = d$), Newton's method converges in a single iteration for all tested d and initial guesses.
- When the multiplicity is incorrect, convergence remains reliable but slows significantly as d increases.
- Steffensen's method consistently fails to satisfy the convergence tolerance for $d \geq 2$, despite iterates drifting toward the root.

Figure (Task 1, Figure 1) shows the growth in iteration counts as d increases, while Figures (Task 1, Figures 2–6) further illustrate sensitivity to both the multiplicity parameter and the initial guess. Overall, the numerical results align closely with the expected theoretical behavior of Newton-type methods at multiple roots.

6.2 Task 2: Basins of attraction, scaling, and root coalescence

Task 2 focuses on global convergence behavior for the cubic

$$f(x) = x(x-\rho)(x+\rho), \qquad \rho = 1,$$

which has simple roots at -1, 0, and 1.
Table 2 summarizes a representative set of demo runs from Task 2.1.

x_0	Iter	x_{end}	Root hit
$+2.0$	7	$+1.0000$	$+1$
$+0.2$	3	≈ 0	0
-0.8	5	-1.0000	-1
-2.0	7	-1.0000	-1

Table 2: Task 2.1: Representative runs illustrating attraction to different roots.

These results reflect the partitioning of the real line into basins of attraction, separated by the critical points $\pm\alpha$ where $f'(x) = 0$. A sweep over 3001 initial values confirms that all valid starting points converge to one of the three roots, with no unexpected failures.

Figures (Task 2, Figures 1–3) visualize these basins and show sharp transitions near the critical points. Tasks 2.2 and 2.3 demonstrate that scaling the function or allowing roots to coalesce preserves the qualitative basin structure, though convergence rates deteriorate near multiple roots.

6.3 Task 3: Scalar nonlinear relaxation and singularities

Task 3 studies the fixed-point iteration

$$\phi(x) = x - \frac{x^2-3}{x^2+2x-3},$$

which has fixed points at $\pm\sqrt{3}$ and singularities at $x = 1$ and $x = -3$.
Table 3 summarizes the five representative runs reported by the driver.

Run	x_0	Iter	x_{end}	Status
1	$+1.83$	4	$+\sqrt{3}$	converged
2	-1.83	4	$-\sqrt{3}$	converged
3	$+0.50$	6	$-\sqrt{3}$	converged
4	$+1.20$	8	$+\sqrt{3}$	converged
5	-3.20	60	diverged	stalled

Table 3: Task 3: Summary of scalar relaxation runs.

The results show that initial conditions near singularities produce large transient excursions, and that iterates may cross between basins of attraction. Starting to the left of the singularity at $x = -3$ leads to divergence or stagnation. Figures (Task 3, Figures 1–4) illustrate the iterate trajectories and residual decay.

6.4 Task 4: Non linear relaxation for systems

Task 4 extends the analysis to a two-dimensional system with two solutions and compares the relaxation mappings G_1 and G_2.

Representative runs are summarized in Table 4.

Method	x_0	Iter	Converged to
G_1	$(0.5, -0.5)$	36	fail
G_1	$(1.2, -1.5)$	35	root_pos
G_2	$(0.5, -0.5)$	22	root_neg
G_2	$(-1.2, 1.2)$	22	root_neg

Table 4: Task 4: Representative nonlinear system relaxation runs.

A full 61×61 grid sweep shows that G_1 converges exclusively to the positive solution when it converges, while G_2 converges exclusively to the negative solution. G_2 also converges faster on average and from a larger set of valid initial conditions. Figures (Task 4, Figures 1–6) visualize the convergence regions and iteration counts.

7 Conclusions

This project examined the behavior of several nonlinear root-finding and fixed-point methods through a sequence of controlled numerical experiments. In **Task 1**, the Newton family clearly illustrates how root multiplicity governs convergence: standard Newton and Steffensen lose quadratic convergence at multiple roots, while the modified Newton method with the correct multiplicity parameter collapses the problem in a single step. The tables and plots show that iteration counts alone can be misleading, since convergence is determined by the decay of the residual rather than proximity in x.

Task 2 highlights the dynamical systems viewpoint of iterative methods. By studying sensitivity to initial conditions, scaling, and root coalescence, we see how basin structure and local geometry influence which root is selected and whether convergence occurs at all. In particular, the numerical experiments confirm that changes in scaling or root separation can significantly alter convergence behavior even when the algebraic form of the iteration is unchanged.

In **Task 3**, the nonlinear relaxation example reinforces the importance of singularities and invariant regions. Initial conditions near poles slow convergence but do not necessarily prevent it, while starting outside the natural basin leads to divergence or stagnation. These observations are consistent with the fixed-point and contraction-mapping analysis developed earlier and emphasize the role of local stability rather than global guarantees.

Finally, **Task 4** demonstrates how these ideas extend to systems. Different fixed-point maps for the same nonlinear system exhibit markedly different robustness properties, basin structures, and convergence rates. The grid-based experiments show that convergence is highly map-dependent: one iteration reliably converges to a single solution branch, while the other converges to the opposite branch and does so more efficiently. Overall, the results emphasize that convergence behavior is driven as much by the choice of iteration map as by the underlying nonlinear problem itself.

8 Disclaimer

ChatGPT was used to help debug and design parts of the driver and to help with a few subroutines as well as to help with some LaTeXformatting in this report. While ChatGPT was heavily used for the driver. I have made sure to add comments along the code to insure I knew what was going on.

A Appendix A: Figures

All figures below were generated by the task drivers and are grouped by task. Multiple related figures are arranged side-by-side to reduce page length.

Task 1: Newton Family and Steffensen

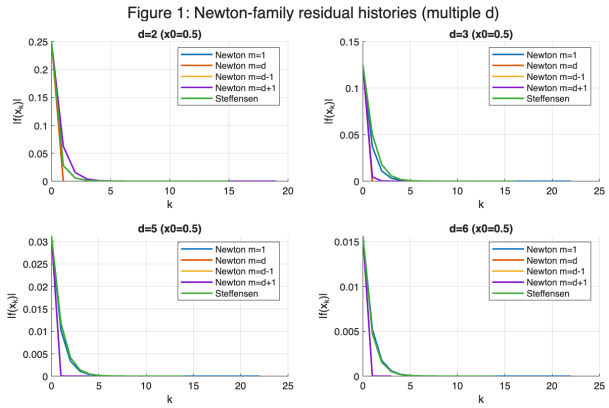

Figure 1: *
(a) Iteration counts vs. multiplicity d.

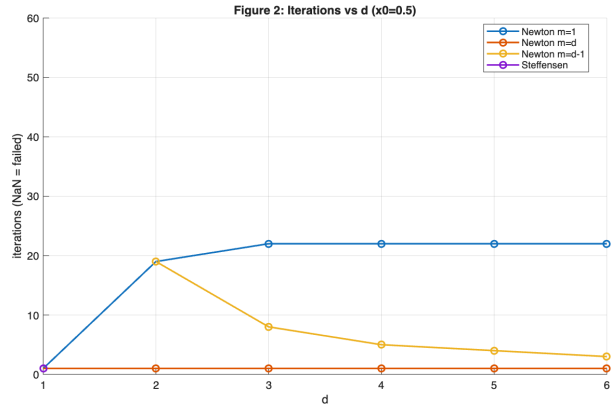

Figure 2: *
(b) Final residuals $|f(x_{\text{end}})|$ vs. d.

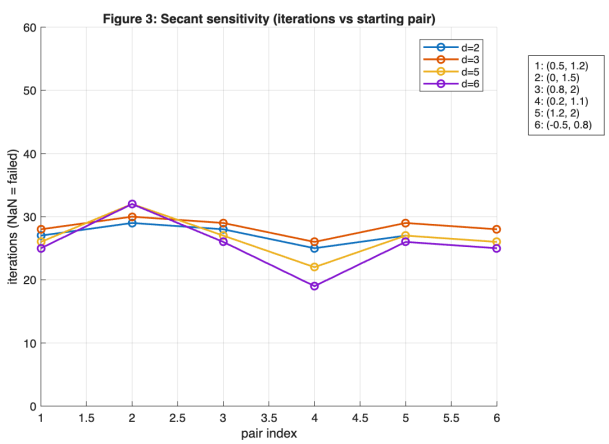

Figure 3: *
(c) Sensitivity to initial conditions.

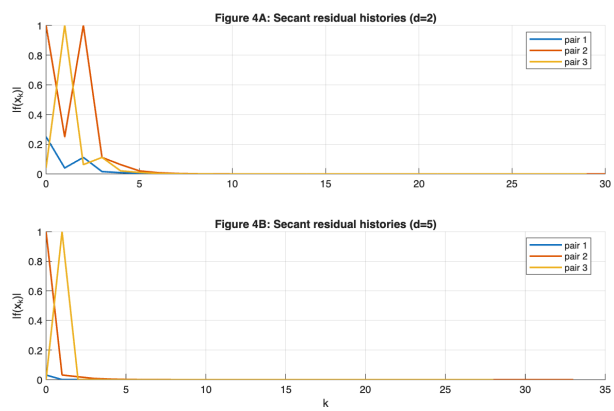

Figure 4: *
(d) Convergence flag outcomes.

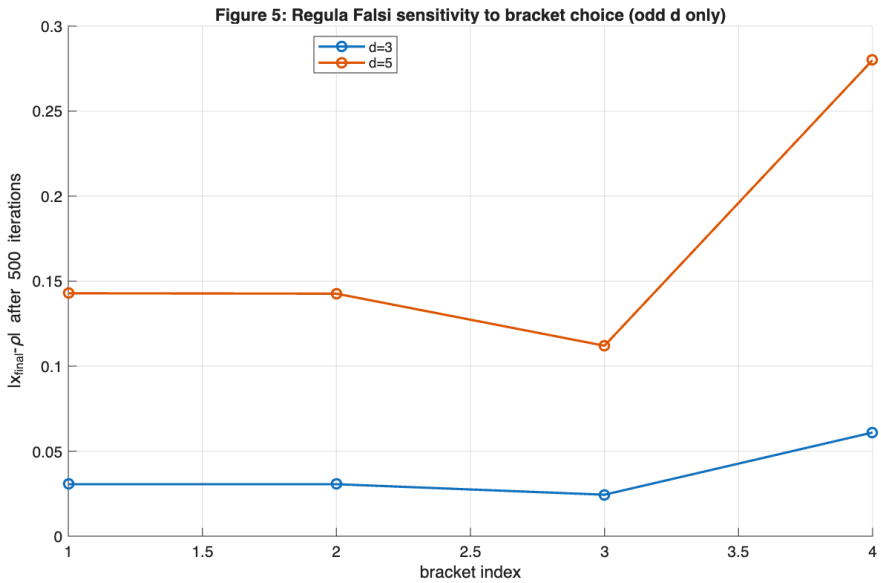

Figure 5: *

(e) Effect of incorrect multiplicity parameter m in modified Newton's method.

Task 2: Sensitivity, Scaling, and Root Coalescence

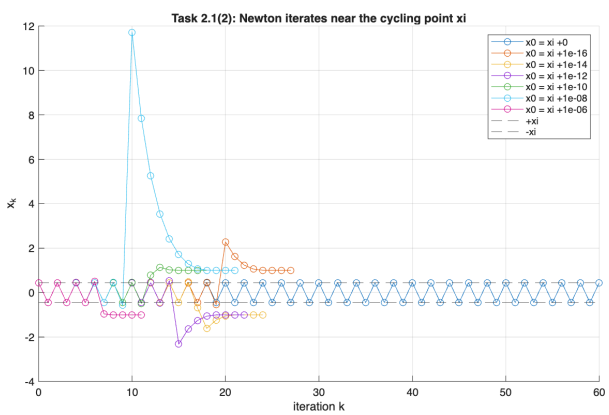

Figure 6: *

(a) Basins of attraction for $f(x) = x(x-\rho)(x+\rho)$.

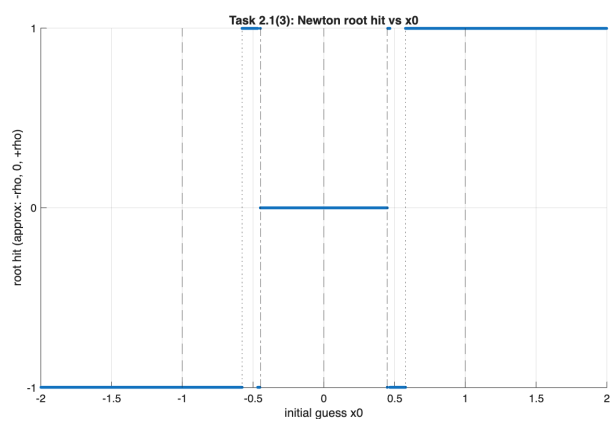

Figure 7: *

(b) Effect of scaling $f \mapsto \sigma f$.

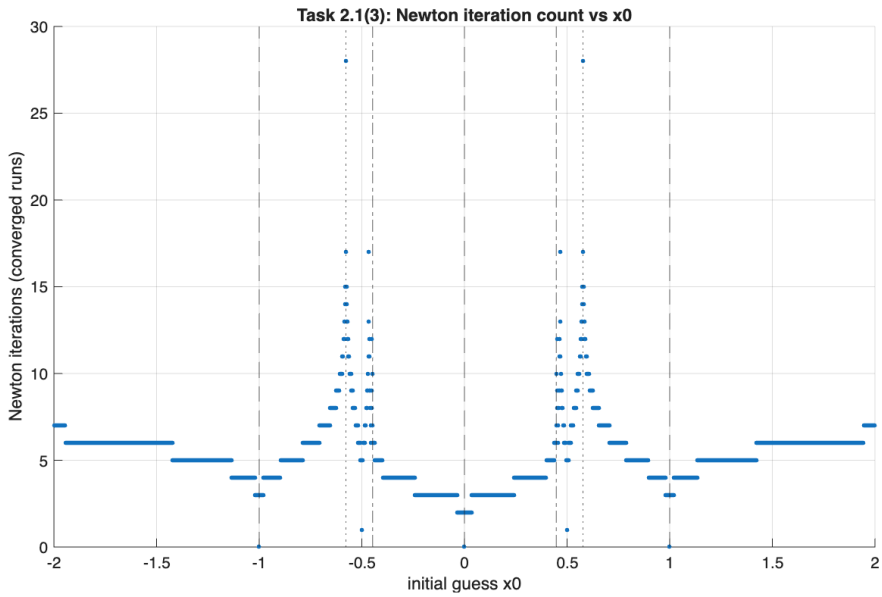

Figure 8: *
(c) Root coalescence and loss of quadratic convergence.

Task 3: Scalar Nonlinear Relaxation

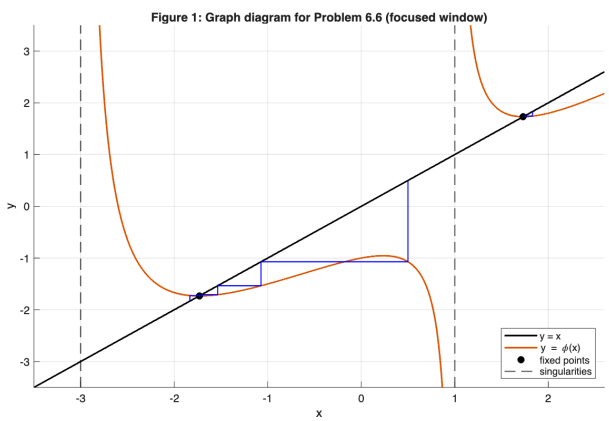

Figure 9: *
(a) Convergence near $\pm\sqrt{3}$.

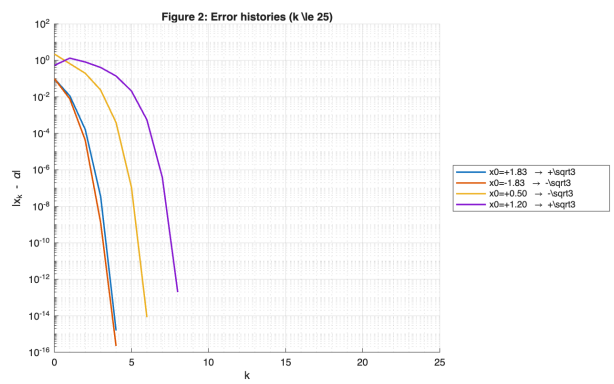

Figure 10: *
(b) Iteration near singularity $x = 1$.

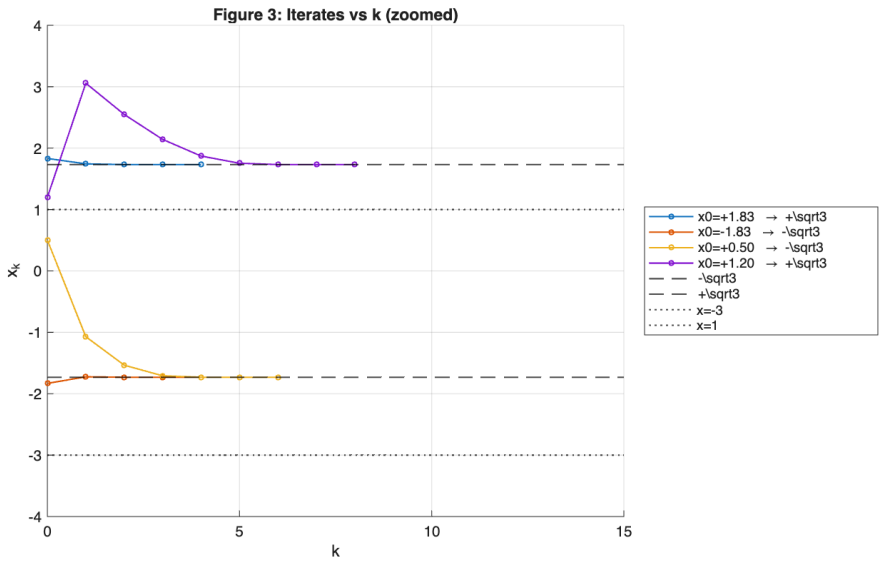

Figure 11: *
(c) Divergence for initial data left of $x = -3$.

Task 4: Nonlinear Relaxation for Systems

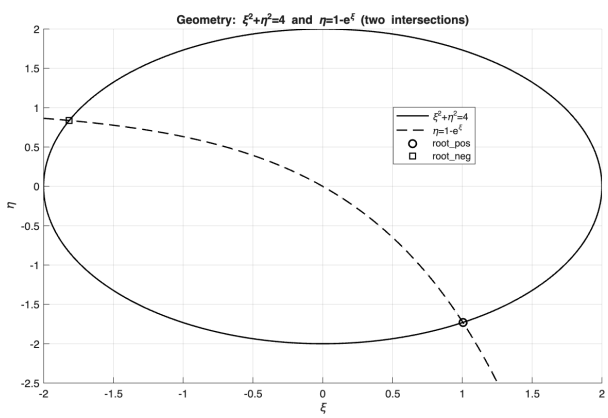

Figure 12: *
(a) Convergence of G_1 to the positive solution branch.

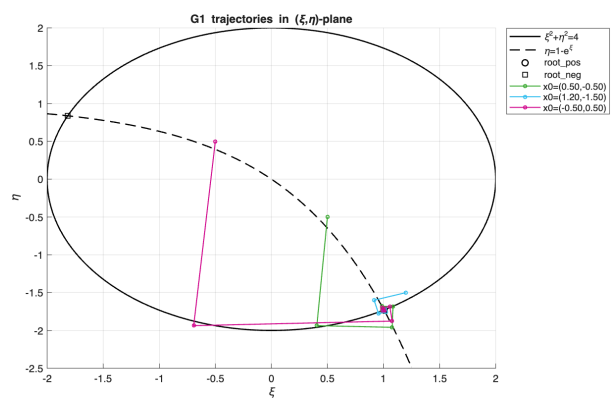

Figure 13: *
(b) Convergence of G_2 to the negative solution branch.

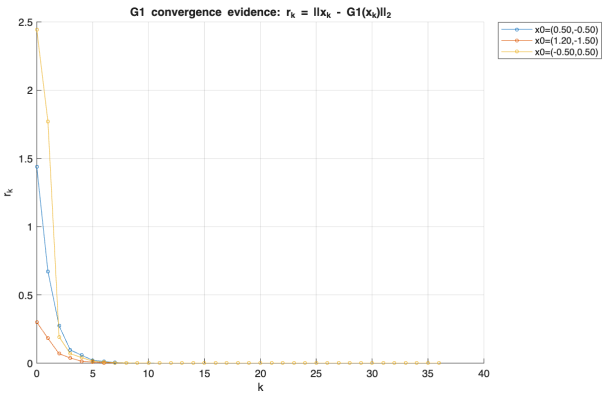

Figure 14: *
(c) Grid robustness for G_1.

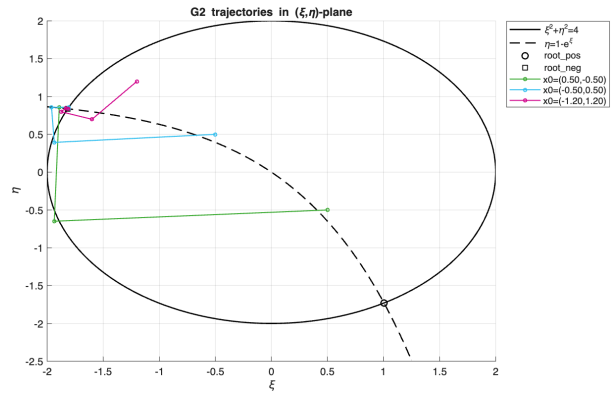

Figure 15: *
(d) Grid robustness for G_2.